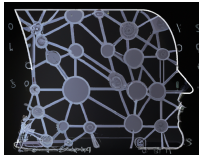
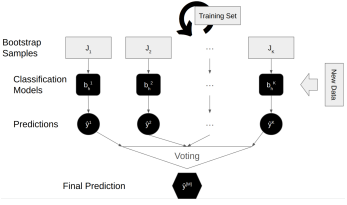
# Applied Machine Learning



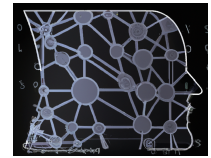
# Ensembling:
# Intro, Model Averaging

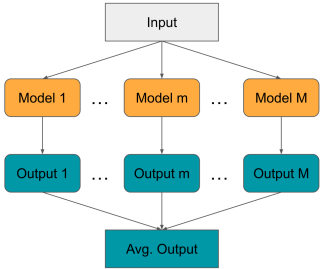



**Learning goals**

- What is Ensembling?
- Bagging and Boosting Recap
- Model Averaging

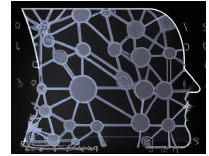
# WHAT IS ENSEMBLING?



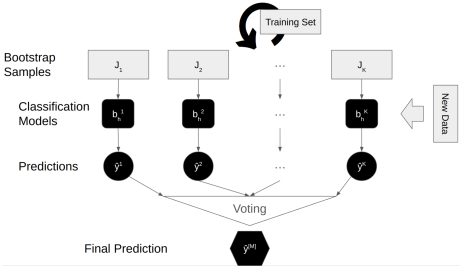



- Combine the predictions of models
- Make weak learners strong
- Make strong learners stronger
- Also: Ensembling effectively *circumvents* the problem of model selection

# RECAP: BAGGING



**B**ootstrap **Agg**regat**ing**: Training multiple instances of the same base model on different subsets of the training data, obtained through random sampling with replacement $\rightarrow$ homogeneous & parallel ensemble



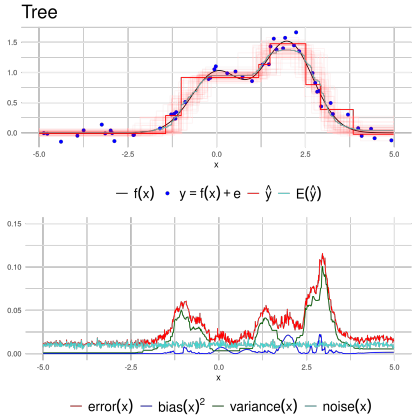

Steps:

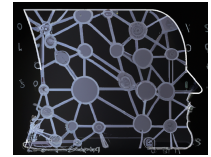
1. Random sampling with replacement to generate different train data sets
2. Base model training
3. Combine predictions

Popular use case: Construction of RF using trees as base models

# RECAP: BAGGING

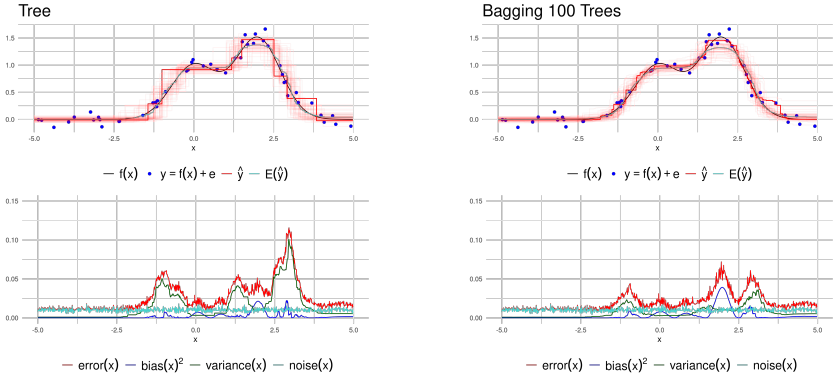
Bias vs. variance: Error = Variance + Bias + Noise



# RECAP: BAGGING

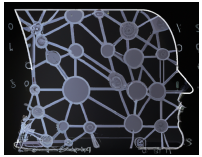
Bias vs. variance bagging: Variance reduction through averaging; Bias of model is slightly increased as less data are seen




Works well for unstable learners

NB: RF also uses col subsampling

# RECAP: BOOSTING

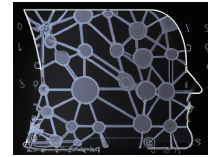


Iterative technique that sequentially trains subsequent base models to correct errors of previous models, for example, by re-weighting instances or predicting pseudo-residuals.

Examples:

- AdaBoost (Adaptive Boosting)
- Gradient Boosting
- XGBoost (Extreme Gradient Boosting)
- LightGBM (fast and memory efficient Gradient Boosting)
- CatBoost (native support for categorical features + ordered boosting)
- Model Based Boosting

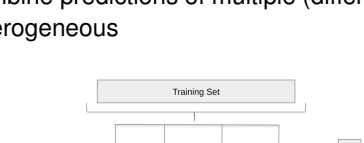
# GENERAL NOTATION FOR ENSEMBLING



- $\{b^{[1]}, \ldots, b^{[M]}\} \subseteq \mathcal{H}_1 \times \ldots \times \mathcal{H}_M$ set of $M$ models trained on training data $\mathcal{D}$
- $b_\pi^{[m]} : \mathcal{X} \to [0,1]^g$ probability predictions under the $m$-th model when doing classification (w.l.o.g. we assume $g = 1$ in the following and $b_\pi^{[m]}$ returning the probabilities of the positive class)
- $b_h^{[m]} : \mathcal{X} \to \mathcal{Y}$ hard label predictions under the $m$-th model when doing classification or regression
- $\hat{f}^{[M]}(\mathbf{x}) = G(b^{[1]}(\mathbf{x}), \ldots, b^{[M]}(\mathbf{x}))$ prediction of an ensemble model obtained via aggregating function $G$

# MODEL AVERAGING

Combine predictions of multiple (different) models to make a final prediction $\rightarrow$ heterogeneous




Steps:

1. Base model training
2. Combine predictions

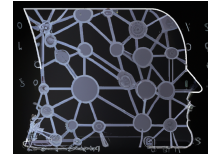
Classification: majority voting (hard voting) / model averaging (soft voting)

- $\hat{f}^{[M]}(\mathbf{x}) = \text{mode}(\{b_h^{[1]}(\mathbf{x}), \ldots, b_h^{[M]}(\mathbf{x})\})$ assuming hard label predictions
- $\hat{f}^{[M]}(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} b_\pi^{[m]}(\mathbf{x})$ assuming probability predictions

Regression: model averaging

- $\hat{f}^{[M]}(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} b_h^{[m]}(\mathbf{x})$

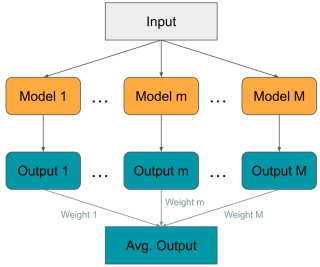
# WEIGHTED MODEL AVERAGING



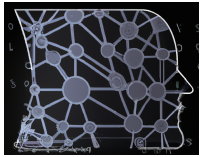
Assume a set of $M$ different models obtained from data $\mathcal{D}$ using a resampling $\mathcal{J}$.

Assign weights $\mathbf{w} = (w_1, \ldots, w_M)$, usually: $0 \leq w_m \leq 1, \sum_{m=1}^{M} w_m = 1$ to the models

$$\hat{f}_{\mathbf{w}}^{[M]}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} w_m \cdot b_{\pi}^{[m]}(\mathbf{x})$$



# EXAMPLE: WEIGHTED MODEL AVERAGING



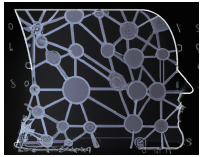
Balanced accuracy (BACC) on the sonar task, using a stratified holdout split (1/3 test set):

| **Learner** | **BACC** (Test) |
|---|---|
| RF | 0.834 |
| k-NN (k=7) | 0.792 |
| Log. Reg. | 0.695 |
| Model Averaging (Equal Weights) | 0.796 |
| Model Averaging ($\mathbf{w} = (0.75, 0.05, 0.20)^\intercal$) | 0.810 |

How can we find 'good' weights?

NB: Model averaging with equal weights will not result in better performance out of the box; depends on types of learners, their diversity in predictions, ...

# WEIGHTED MODEL AVERAGING CONTINUED



Find the optimal weight vector $\mathbf{w}^*$ such that the estimated generalization error (based on a performance metric $\rho$ or loss function $L$) of the ensemble is optimal:

$$\mathbf{w}^* := \arg\min_{\mathbf{w}\in\mathbb{R}^M} \hat{\mathrm{GE}}\left(\hat{f}_{\mathbf{w}}, \mathcal{J}, \rho\right)$$

Performance metric/loss function unknown a priori (e.g., as in AutoML) $\rightarrow$ black box optimization problem.

Difficulty of the optimization problem depends on the metric/loss function:

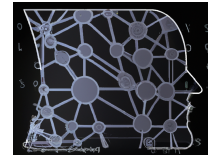
- MSE (regression) $L_{\mathrm{MSE}}(y^{(i)}, \hat{f}_{\mathbf{w}}(\mathbf{x}^{(i)})) = (y^{(i)} - \hat{f}_{\mathbf{w}}(\mathbf{x}^{(i)}))^2$;
  Constraints of the form $0 \leq w_m \leq 1, \sum_{m=1}^{M} w_m = 1$;

  $$\arg\min_{\mathbf{w}\in\mathbb{R}^M} \hat{\mathrm{GE}}(\hat{f}_{\mathbf{w}}, \mathcal{J}, L_{\mathrm{MSE}}) \text{ s.t. } 0 \leq w_m \leq 1, \sum_{m=1}^{M} w_m = 1$$

  can be written as a quadratic program.
- 0-1 loss (binary classification); the optimization problem is NP-hard.

In practice, we use black box optimization, greedy ensemble selection, or a linear model (stacking).

# GREEDY ENSEMBLE SELECTION



Assume a set of $M$ different models obtained from data $\mathcal{D}$ using a resampling $\mathcal{J}$.
We now want to optimize the weights used in weighted model averaging.

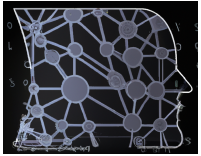
Greedy ensemble selection (GES, see "Caruana et al." 2004) is one of the most popular methods to do so:

1. Start with the empty ensemble
2. Add the base model that maximizes the ensembles' performance metric on a hillclimb (validation) set
3. Repeat step 2 until a given number of iterations are reached
4. Return the ensemble from the set of generated ensembles that has the best performance on the validation set

NB: GES optimizes the weights on an integer domain (i.e., base models are selected one or multiple times or not at all[1].)

[1] This has the (positive) side-effect of resulting in a sparse weight vector

# EXAMPLE: GES



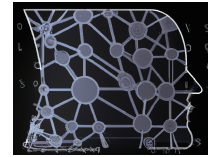
| Iter | Sel. Learners | Current BACC (Valid) | BACC if Learner added (Valid) | | |
|---|---|---|---|---|---|
| | | | RF | k-NN | Log. Reg. |
| 1 | {RF: 0, k-NN: 0, Log. Reg.: 0} | NA | **0.837** | 0.809 | 0.766 |
| 2 | {RF: 1, k-NN: 0, Log. Reg.: 0} | 0.837 | **0.837** | 0.809 | 0.766 |
| 3 | {RF: 2, k-NN: 0, Log. Reg.: 0} | 0.837 | **0.837** | 0.809 | 0.809 |
| 4 | {RF: 3, k-NN: 0, Log. Reg.: 0} | 0.837 | **0.837** | 0.769 | 0.832 |
| 5 | {RF: 4, k-NN: 0, Log. Reg.: 0} | 0.837 | 0.837 | 0.792 | **0.855** |
| 6 | {RF: 4, k-NN: 0, Log. Reg.: 1} | 0.855 | 0.897 | **0.917** | 0.809 |
| 7 | {RF: 4, k-NN: 1, Log. Reg.: 1} | **0.917** | **0.917** | 0.895 | 0.809 |
| 8 | {RF: 5, k-NN: 1, Log. Reg.: 1} | 0.917 | **0.897** | 0.875 | 0.875 |
| 9 | {RF: 6, k-NN: 1, Log. Reg.: 1} | 0.897 | **0.877** | 0.875 | 0.875 |
| 10 | {RF: 7, k-NN: 1, Log. Reg.: 1} | 0.877 | 0.857 | 0.855 | **0.917** |
| 11 | {RF: 7, k-NN: 1, Log. Reg.: 2} | 0.917 | ... | ... | ... |

Best iteration: 6 with BACC (Valid) of 0.917

$\rightarrow$ {RF: 4, k-NN: 1, Log. Reg.: 1}

$\rightarrow \mathbf{w}^* = (\frac{4}{6}, \frac{1}{6}, \frac{1}{6})^\intercal$

# EXAMPLE: CONCLUSION



| **Learner** | **BACC** (Valid) | **BACC** (Test) |
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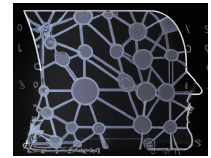
| Model Averaging (Equal Weights) | 0.832 | 0.796 |
| Model Averaging ($\mathbf{w} = (0.75, 0.05, 0.20)^\intercal$) | 0.875 | 0.810 |
| Model Averaging (GES: $\mathbf{w}^* = (\frac{4}{6}, \frac{1}{6}, \frac{1}{6})^\intercal$) | 0.917 | 0.864 |

NB: Optimizing weights can (and often does) overfit to the validation data; especially if standard black box optimizers are used

# CLOSING



Why does ensembling work? ▸ “Dietterich” 2000

- Statistical Problem
- Computational Problem
- Representational Problem

Requirements to build an ensemble:

- Diverse Models
- Strong Models

Practical Suggestion: Greedy Ensemble Selection works really well in practice.